

True/False (5 points):

7.5



ILLINOIS INSTITUTE OF TECHNOLOGY

Department of Computer Science

## CS 458 Final Exam

May 2025

Name: \_\_\_\_\_

Student ID: A. \_\_\_\_\_

1. Substitution ciphers are vulnerable to frequency analysis in the plaintext. (True / False) ☒ True
2. The Feistel cipher structure is used in both DES and AES. (True / False) ☒ False
3. ECB mode encrypts identical plaintext blocks into identical ciphertext blocks. (True / False) ☒ False
4. The MixColumns operation in AES provides diffusion by mixing the bytes in each column of the state. (True / False) ☒ True
5. Public-key cryptosystems are generally faster than symmetric key systems. (True / False) ☒ False
6. The S-boxes in DES are responsible for the diffusion property of the cipher. (True / False) ☒ False
7. In the Feistel structure, the same algorithm is used for both encryption and decryption. (True / False) ☒ True
8. A major advantage of public-key cryptography is that it eliminates the key distribution and digital signature problems present in symmetric cryptography. (True / False) ☒ False
9. Digital signatures provide non-repudiation, ensuring that the sender cannot deny having sent the message. (True / False) ☒ True
10. The Diffie-Hellman key exchange algorithm can be used for both key exchange and digital signatures. (True / False) ☒ False
11. Digital signatures can provide message confidentiality. (True / False) ☒ False
12. Non-repudiation ensures that the sender of a message can deny the creation of the message. (True / False) ☒ False
13. A secure cryptographic hash function should be deterministic, meaning the same input data should always produce the same hash output. (True / False) ☒ True
14. It is computationally feasible to find any pairs of inputs  $x_1 \neq x_2$  such that  $H(x_1) = H(x_2)$  in a secure cryptographic hash function. (True / False) ☒ False
15. The SHA-256 hash function produces a 256-bit hash value. (True / False) ☒ True
16. Cryptographic hash functions are designed to be easily reversible. (True / False) ☒ False
17. Padding schemes for digital signatures are primarily used to prevent attacks such as existential forgery, making it more difficult for attackers to generate valid message-signature pairs. (True / False) ☒ True

7.5  
9

6  
10

21  
21

$\Sigma = 34.5$

True/False (9 points):

7.5

1. Substitution ciphers work by rearranging the positions of letters in the plaintext. (True / False)
2. The Feistel cipher structure is used in both DES and AES. (True / False)
3. ECB mode encrypts identical plaintext blocks into identical ciphertext blocks. (True / False)
4. The MixColumns operation in AES provides diffusion by mixing the bytes in each column of the state. (True / False)
5. Public-key cryptosystems are generally faster than symmetric key systems. (True / False)
6. The S-boxes in DES are responsible for the diffusion property of the cipher. (True / False)
7. In the Feistel structure, the same algorithm is used for both encryption and decryption. (True / False)
8. A major advantage of public-key cryptography is that it eliminates the key distribution and digital signature problems present in symmetric cryptography. (True / False)
9. Digital signatures provide non-repudiation, ensuring that the sender cannot deny having sent the message. (True / False)
10. The Diffie-Hellman key exchange algorithm can be used for both key exchange and digital signatures. (True / False)
11. Digital signatures can provide message confidentiality. (True / False)
12. Non-repudiation ensures that the sender of a message can deny the creation of the message. (True / False)
13. Digital signatures are more difficult to duplicate or tamper with than conventional signatures. (True / False)
14. A cryptographic hash function should be deterministic, meaning the same input data should always produce the same hash output. (True / False)
15. It is computationally feasible to find any pairs of inputs  $x_1 \neq x_2$  such that  $H(x_1) = H(x_2)$  in a secure cryptographic hash function. (True / False)
16. The SHA-256 hash function produces a 512-bit hash value. (True / False)
17. Cryptographic hash functions are designed to be easily reversible. (True / False)
18. Padding schemes in digital signatures are primarily used to prevent attacks such as existential forgery, making it more difficult for attackers to generate valid message-signature pairs. (True / False)

**Multiple choice (10 points):**

1. How many rounds does the AES algorithm use when the key length is 128 bits?
  - A) 10
  - **B) 12**
  - C) 14
  - D) 16
2. What does the S-Box in the AES algorithm provide?
  - A) Key generation
  - **B) Non-linear substitution**
  - C) Permutation
  - D) Linear transformation
3. The security of RSA is based on the difficulty of which mathematical problem?
  - **A) Discrete logarithm problem**
  - B) Factoring large integers
  - C) Matrix multiplication
  - D) Polynomial interpolation
4. What is the main advantage of Elliptic Curve Cryptography (ECC) over RSA?
  - A) Encryption Algorithm
  - **B) Smaller key size for the same security level**
  - C) Easier key management
  - D) Simpler implementation
5. In the context of Diffie-Hellman key exchange, what is the primary mathematical problem that ensures its security?
  - A) Integer factorization problem
  - B) Discrete logarithm problem
  - **C) Elliptic curve problem**
  - D) RSA problem
6. Which of the following is not a core security service provided by digital signatures?
  - A) Message Integrity
  - **B) Message Confidentiality**
  - C) Message Authentication
  - D) Non-repudiation

7. What cryptographic method fails to provide a desirable security function in the example of Bob ordering a pink car?

- A) Symmetric cryptography
- B) Asymmetric cryptography
- C) Public Key Infrastructure (PKI)
- D) Digital certificates

8. What ensures that a message has not been modified in transit?

- A) Confidentiality
- B) Message Integrity
- C) Non-repudiation
- D) Authentication

9. Which digital signature scheme is known for verifying a message by decrypting the signature with the signer's public key?

- A) ElGamal Digital Signature Scheme
- B) Digital Signature Standard (DSS)
- C) Schnorr Digital Signature Scheme
- D) RSA Digital Signature Scheme

10. Which property ensures that it is infeasible to find two different inputs that produce the same hash output?

- a) Pre-image resistance
- b) Collision resistance
- c) Second pre-image resistance
- d) Deterministic property

3, 5, 7, 11, 13, 17, 19, 23, 29

**Break The RSA and produce Digital Signature (21 points):**

We have an RSA public-key cryptosystem between Alice and Bob.

Alice's public key:  $(N=33, e=3)$

Bob's Public key:  $(N=21, e=5)$

Bob's tasks:

1. (7 points) Break the system and find  $P, q, \phi(N)$ , private key  $d$ . For both Alice and Bob (Show your work in detail and write all formulas)

Alice:  $e \cdot d = 1 \pmod{\phi(N)}$

$$3 \cdot d = 1 \pmod{20}$$

$$\boxed{d=7}$$

$$3 \cdot 7 = 21 - 20 = 1$$

$$\begin{aligned}\phi(N) &= (p-1)(q-1) \\ &= (2)(10) \\ \phi(N) &= 20\end{aligned}$$

$$\begin{aligned}N &= p \times q \\ 33 &= 3 \times 11 \\ \boxed{p=3} \\ \boxed{q=11}\end{aligned}$$

5. Encrypt signature value and send it to Alice ( $C$  = Encrypted Signature Value)

Bob:  $e \cdot d = 1 \pmod{\phi(N)}$

$$5 \cdot d = 1 \pmod{12}$$

$$5 \cdot 5 = 25 \pmod{12} \equiv 1$$

$$\boxed{d=5}$$

$$\begin{aligned}\phi(N) &= (p-1)(q-1) \\ &= 2 \cdot 6 \\ \phi(N) &= 12\end{aligned}$$

$$\begin{aligned}N &= p \times q \\ 21 &= 3 \times 7 \\ \boxed{p=3} \\ \boxed{q=7}\end{aligned}$$

5 = 101 in binary

$$\begin{array}{r} 2 \quad 4 \times 4 = 16 \\ 3 \quad 16 \times 4 = 64 \\ 4 \quad 64 \times 4 = 256 \\ \times 4 \\ \hline 256 \end{array} \quad \begin{array}{r} 5 \quad 256 \\ \times 4 \\ \hline 1024 \end{array}$$

2. (14 points) Bob wants to sign a message ( $M = 4$ ), then encrypt it and send it to Alice. Explain the whole process and calculations, and find all the numbers, such as: (Show your work in detail and write all formulas)

$$e_B = 5 \quad N_B = 21 \quad d_B = 5$$

$$e_A = 3 \quad N_A = 33 \quad d_A = 7$$

$$\begin{array}{r} 48 \\ 21 \overline{) 1024} \\ \underline{84} \phantom{00} \\ 184 \phantom{0} \\ \underline{168} \phantom{0} \\ 160 \\ \underline{147} \\ 13 \end{array}$$

Bob's tasks:

a. Sign a message ( $M = 4$ ) and calculate the signature value ( $S = \text{Signature Value}$ )

$$S = M^{d_B} \bmod N_B$$

$$4^5 \bmod 21$$

|       |          |                      |
|-------|----------|----------------------|
|       | <u>E</u> | <u>R</u>             |
| $1^1$ | $4^1$    | 4                    |
| $1^2$ | $4^2$    | 16                   |
| $1^4$ | $16^2$   | $256 \bmod 21 = 4$   |
|       |          | $4 \cdot 4 \bmod 21$ |
|       |          | $16 \bmod 21 = 16$   |

$$\boxed{S = 16}$$

$$\begin{array}{r} 21 \\ \times 12 \\ \hline 42 \\ + 210 \\ \hline 252 \end{array}$$

$$\begin{array}{r} 16 \\ 16 \\ \hline 96 \\ 160 \\ \hline 256 \end{array}$$

b. Encrypt signature value and send it to Alice ( $C = \text{Encrypted Signature Value}$ )

$$C = S^{e_A} \bmod N_A$$

$$16^3 \bmod 33$$

3 = 11 in binary

|       |          |                     |
|-------|----------|---------------------|
|       | <u>E</u> | <u>R</u>            |
| $1^1$ | 16       | 16                  |
| $1^2$ | $16^2$   | $256 \bmod 33 = 25$ |

$$16 \cdot 25 \bmod 33$$

$$400 \bmod 33 = 4$$

$$\boxed{C = 4}$$

$$\begin{array}{r} 233 \\ \times 7 \\ \hline 231 \end{array}$$

$$\begin{array}{r} 256 \\ 231 \\ \hline 25 \end{array}$$

$$\begin{array}{r} 25 \\ \times 16 \\ \hline 150 \\ 250 \\ \hline 400 \end{array}$$

$$\begin{array}{r} 33 \\ \times 12 \\ \hline 66 \\ 330 \\ \hline 396 \end{array}$$

$$e_B = 5 \quad N_B = 21 \quad d_B = 5 \quad C = 4$$

$$e_A = 3 \quad N_A = 33 \quad d_A = 7$$

Alice's tasks:

7 = 111 in binary

c. Decrypt the received encrypted signature ( $S' = \text{Decrypted Signature Value}$ )

$$S' = C^{d_A} \bmod N_A$$

$$= 4^7 \bmod 33$$

|    |                 |                          |                          |
|----|-----------------|--------------------------|--------------------------|
|    | $\frac{E}{1^2}$ | $\frac{R}{4}$            |                          |
| 12 | $4^1$           | 4                        |                          |
| 12 | $4^2$           | 16                       |                          |
| 14 | $16^2$          | $256 \bmod 33 \equiv 25$ | $256 \bmod 33 \equiv 25$ |

$$(256 \cdot 4 \cdot 16) \bmod 33$$

$$25 \cdot 4 \cdot 16 \bmod 33$$

$$100 \bmod 33 \equiv 1$$

$$1 \cdot 16 \bmod 33 \equiv 16$$

$$\boxed{S' = 16}$$

d. Verify the signature and signer identity ( $M' = \text{Extracted Message Value}$ )

$$M' = S'^{e_B} \bmod N_B$$

5 = 101 in binary

$$16^5 \bmod 21$$

|    |                 |                         |
|----|-----------------|-------------------------|
|    | $\frac{E}{1^1}$ | $\frac{R}{16}$          |
| 12 | $16^1$          | 16                      |
| 12 | $16^2$          | $256 \bmod 21 \equiv 4$ |
| 14 | $4^2$           | 16                      |

$$16 \cdot 16 \bmod 21$$

$$256 \bmod 21 \equiv 4$$

$$\boxed{M' = 4}$$



ILLINOIS INSTITUTE OF TECHNOLOGY

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## CS 458 Midterm Exam

Spring 2025

Name: \_\_\_\_\_

Student ID: \_\_\_\_\_

32 4 11  
35 12 15

Grade (out of 62): 47

17

1. True/False (20 points):

1. Replay attacks are a type of passive attack. (True / False)
2. Substitution ciphers work by rearranging the positions of letters in the plaintext. (True / False)
3. In a Vigenère cipher, the same substitution is applied to each letter of the message. (True / False)
4. The Feistel cipher structure is used in both DES and AES. (True / False)
5. The primary weakness of DES is its small key size. (True / False)
6. ECB mode encrypts identical plaintext blocks into identical ciphertext blocks. (True / False)
7. Information assurance involves protecting the confidentiality, integrity, and availability of information. (True / False)
8. The MixColumns operation in AES provides diffusion by mixing the bytes in each column of the state. (True / False)
9. The initial and final permutations in DES have significant cryptographic importance. (True / False)
10. Public-key cryptosystems are generally faster than symmetric key systems. (True / False)
11. In symmetric cryptography, a stream cipher encrypts data in fixed-size blocks. (True / False)
12. The S-boxes in DES are responsible for the diffusion property of the cipher. (True / False)
13. The Counter (CTR) mode of operation allows block ciphers to function like stream ciphers. (True / False)
14. In the Feistel structure, the same algorithm is used for both encryption and decryption. (True / False)
15. The Avalanche effect in DES ensures that a small change in plaintext results in a significant change in ciphertext. (True / False)
16. A major advantage of public-key cryptography is that it eliminates the key distribution and digital signature problems present in symmetric cryptography. (True / False)
17. The security of RSA is based on the difficulty of factoring large integers. (True / False)
18. Digital signatures provide non-repudiation, ensuring that the sender cannot deny having sent the message. (True / False)
19. Padding is used in RSA to prevent deterministic encryption. (True / False)
20. The Diffie-Hellman key exchange algorithm can be used for both key exchange and digital signatures. (True / False)

2. Multiple choice (15 points):

15

1. Which principle states that a cryptographic system should remain secure even if everything about the system, except the key, is public knowledge?
  - ☒ A) Kerckhoffs' Principle
  - ☐ B) Shannon's Principle
  - ☐ C) Caesar's Principle
  - ☐ D) Diffusion Principle
2. In a symmetric cryptosystem, what is true about the keys used for encryption and decryption?
  - ☐ A) They are different
  - ☒ B) Only one key is used for both encryption and decryption
  - ☐ C) They are always public
  - ☐ D) They are generated randomly each time
3. What type of cipher is the Caesar cipher?
  - ☐ A) Transposition cipher
  - ☐ B) Polyalphabetic cipher
  - ☒ C) Mono-alphabetic substitution cipher
  - ☐ D) Stream cipher
4. Which of the following algorithms is used for key exchange but not for encryption?
  - ☐ A) RSA
  - ☐ B) AES
  - ☒ C) Diffie-Hellman
  - ☐ D) ElGamal
5. What is the main advantage of using the one-time pad cipher?
  - ☐ A) It is very easy to use
  - ☒ B) It is provably secure if used correctly
  - ☐ C) It uses very short keys
  - ☐ D) It requires no key management
6. Which of the following best describes the concept of 'diffusion' in cryptography?
  - ☐ A) Making the relationship between the ciphertext and the key as complex as possible
  - ☒ B) Spreading the influence of each plaintext element over many ciphertext elements
  - ☐ C) Encrypting each character of the plaintext with a different key
  - ☐ D) Ensuring that the same key is not used more than once

7. Which type of cipher rearranges the characters of the plaintext without changing the actual characters used?

- A) Substitution cipher
- **B) Transposition cipher**
- C) Stream cipher
- D) Block cipher

8. In the context of cryptography, what is 'confusion'?

- A) The process of spreading the plaintext over the ciphertext
- **B) The complexity of the relationship between the plaintext and the ciphertext**
- C) The mixing of multiple ciphers
- D) The secret transmission of the key

9. In the context of symmetric encryption, what is a stream cipher?

- A) A cipher that encrypts data in fixed-size blocks
- **B) A cipher that encrypts data one bit or one byte at a time**
- C) A cipher that uses asymmetric keys
- D) A cipher that uses multiple encryption rounds

10. What is the block size used by the AES algorithm?

- A) 64 bits
- **B) 128 bits**
- C) 192 bits
- D) 256 bits

11. How many rounds does the AES algorithm use when the key length is 128 bits?

- **A) 10**
- B) 12
- C) 14
- D) 16

12. What does the S-Box in the AES algorithm provide?

- A) Key generation
- **B) Non-linear substitution**
- C) Permutation
- D) Linear transformation

13. The security of RSA is based on the difficulty of which mathematical problem?

- A) Discrete logarithm problem
- **B) Factoring large integers**
- C) Matrix multiplication
- D) Polynomial interpolation

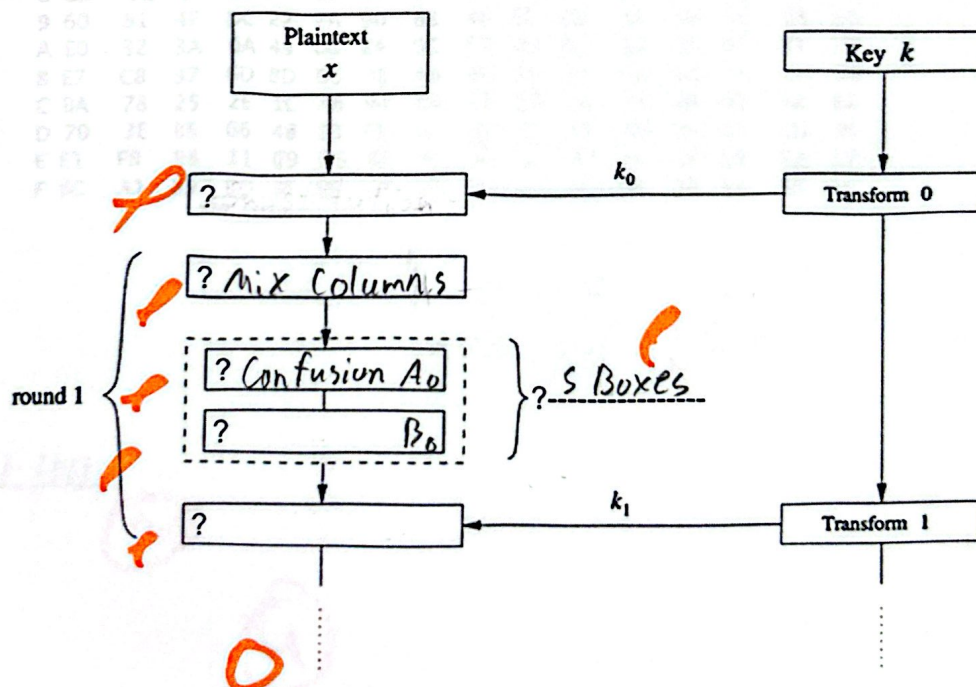
14. What is the main advantage of Elliptic Curve Cryptography (ECC) over RSA?

- A) Encryption Algorithm
- B) Smaller key size for the same security level
- C) Easier key management
- D) Simpler implementation

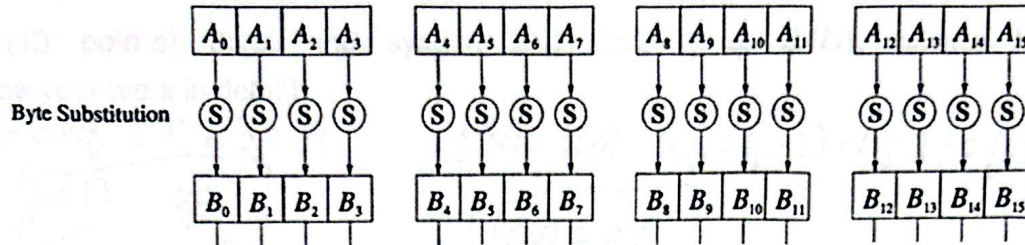
15. In the context of Diffie-Hellman key exchange, what is the primary mathematical problem that ensures its security?

- A) Integer factorization problem
- B) Discrete logarithm problem
- C) Elliptic curve problem
- D) RSA problem

3. AES Structure: Fill in the six blank boxes marked with question marks (6 points)



4. AES S-Box: Substitution values in hexadecimal notation for input byte (xy). (6 points)



Use the given AES S-box table to calculate B0 if A0 : 01101110 (Show you work in detail and give B0 in binary format like A0)

|   | 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | A  | B  | C  | D  | E  | F  |
|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 0 | 63 | 7C | 77 | 7B | F2 | 6B | 6F | C5 | 30 | 01 | 67 | 2B | FE | D7 | AB | 76 |
| 1 | CA | 82 | C9 | 7D | FA | 59 | 47 | F0 | AD | D4 | A2 | AF | 9C | A4 | 72 | C0 |
| 2 | B7 | FD | 93 | 26 | 36 | 3F | F7 | CC | 34 | A5 | E5 | F1 | 71 | D8 | 31 | 15 |
| 3 | 04 | C7 | 23 | C3 | 18 | 96 | 05 | 9A | 07 | 12 | 80 | E2 | EB | 27 | B2 | 75 |
| 4 | 09 | 83 | 2C | 1A | 1B | 6E | 5A | A0 | 52 | 3B | D6 | B3 | 29 | E3 | 2F | 84 |
| 5 | 53 | D1 | 00 | ED | 20 | FC | B1 | 5B | 6A | CB | BE | 39 | 4A | 4C | 58 | CF |
| 6 | D0 | EF | AA | FB | 43 | 4D | 33 | 85 | 45 | F9 | 02 | 7F | 50 | 3C | 9F | A8 |
| 7 | 51 | A3 | 40 | 8F | 92 | 9D | 38 | F5 | BC | B6 | DA | 21 | 10 | FF | F3 | D2 |
| 8 | CD | 0C | 3  | EC | 5F | 97 | 44 | 17 | C4 | A7 | 7E | 3D | 64 | 5D | 19 | 73 |
| 9 | 60 | 81 | 4F | DC | 22 | 2A | 90 | 88 | 46 | EE | B8 | 14 | DE | 5E | 0B | DB |
| A | E0 | 32 | 3A | 0A | 49 | 06 | 24 | 5C | C2 | D3 | AC | 62 | 91 | 95 | E4 | 79 |
| B | E7 | C8 | 37 | 6D | 8D | D5 | 4E | A9 | 6C | 56 | F4 | EA | 65 | 7A | AE | 08 |
| C | BA | 78 | 25 | 2E | 1C | A6 | B4 | C6 | E8 | DD | 74 | 1F | 4B | BD | 8B | 8A |
| D | 70 | 3E | B5 | 66 | 48 | 03 | F6 | 0E | 61 | 35 | 57 | B9 | 86 | C1 | 1D | 9E |
| E | E1 | F8 | 98 | 11 | 69 | D9 | 8E | 94 | 9B | 1E | 87 | E9 | CE | 55 | 28 | DF |
| F | 8C | A1 | 89 | 0D | BF | E6 | 42 | 68 | 41 | 99 | 2D | 0F | B0 | 54 | BB | 16 |

9 F  
1001 1111

B0: 1001 1111

### 5. Break The RSA (15 points):

We have an RSA public-key cryptosystem with a public key ( $N=39, e=5$ ).

1. (10 points) Break the system and find:  $P, q, \phi(N)$ , private key  $d$ .  
(Show your work in detail)

$$N = p \times q = 13 \times 3 = 39$$

$$\boxed{p=13 \quad q=3}$$

$$\phi(N) = \phi(p) \times \phi(q) = (p-1) \times (q-1) = (13-1) \times (3-1) = 12 \times 2 = 24$$

$$\boxed{\phi(N) = 24}$$

$$d \in [1, \phi(N)-1] \rightarrow [1, 23] \text{ which satisfies } e \times d \bmod \phi(N) = 1$$

$$\boxed{d=5}$$

$$5 \times d \bmod 24 = 1$$

$$5 \times 5 = 25 \bmod 24 = 1 \text{ R } 1$$

(10)

2. (5 points) We have a cipher-text:  $C=3$ . Recover corresponding plain-text  $M$  from this cipher-text using your calculated private key. (Show your work)

$$C=3 \quad d=5 \quad C^d$$

$$3^5 = 243 = m?$$

(11)

$$\begin{array}{r} 3 \quad 2 \\ \times 3 \\ \hline 9 \quad 3 \\ 228 \quad 4 \\ \times 3 \\ \hline 81 \quad 5 \\ \times 3 \\ \hline 243 \end{array}$$